Universities Offering Courses In GIS And Related Areas

- Institute of Remote Sensing-Anna Univ.
- Faculty of Science-Banaras Hindu Univ.
- Indian Institute of Remote Sensing (National Remote Sensing Agency)
- Dept. of Applied Sciences,Environmental Science and Engineering Group/Dept. of Remote Sensing-Birla Institute of Technology
- The Centre for Environmental Planning and Technology (CEPT)-Ahmedabad
- GIS Institute, Noida
- Centre of Studies in Resources Engineering/Dept. of Earth Sciences-IIT Mumbai
- Dept. of Civil Engg-IIT Kanpur
- Centre for Atmospheric and Oceanic Sciences-Indian Institute of Science
- Indian School of Mines Dhanbad
- Dept. of Environmental Science-Pune Univ.
- National Remote Sensing Agency
- Dept. of Geography-Univ. of Madras
- Dept. of Space Sciences-Pune Univ.
- Dept. of Earth Sciences-Roorkee Univ.
- Dept. of Geography-Gujarat University
- Dept. of Geography-Jamia Millia Islamia
- School of Environmental Sciences-Jawaharlal Nehru University
- National Bureau of Soil Survey and Land use Planning, Nagpur
- Space Application Center, Ahmedabad
- Geological Survey of India Training Institute, Hyderabad
- NBSS & LUP, Nagpur
- Survey Training Institute, Hyderabad
- Centre of Studies in Resource Engineering, Mumbai
- Dept. of Geography-Utkal Univ.
- Regional Remote Sensing Service Centres at Kharagpur, Nagpur, Dehradun, Jodhpur, Bangalore

lot of accuracy—the Earth is round, maps are flat. Moreover, the special features of the area under consideration also come into play.

Employment opportunities range from organisations developing GIS software solutions, companies providing GIS services, and different departments with the Government.

The Situation

A typical GIS project would be something like developing roads in a village to connect it to the nearest town. There's a lot of work involved, and there are many roles that need playing:

- **Data builders**, well, collect data about the area—temperature, soil quality and other geographical information. This data comes from photographs of the location and images taken by remote sensing, and then compiled into maps, tables and figures.
- **Digitisers** convert all that data to a digital format, which can then be used in software and databases.
- **GIS Developers** use GIS software applications to model the area based on the data, and customise it based on the needs of the project.
- **GIS Analysts** use the GIS data and the original data to check for possible challenges in the project and develop a plan for its execution.
- **The GIS Manager** is in charge of the overall planning, control and management of all activities—from data collection to data analysis.

"The beauty GIS as a career is that it absorbs enthusiasts with both core and non-core GIS skills. Individuals can be recruited by organisations that create GIS products, or those who use these products and offer GIS services. However, for any industry, the main problem lies in creating the data related to our country's demography, geography and infrastructure—this isn't tailor-made and thus needs lot of effort to build", says Manideep Saha, Head-Infrastructure for India and SAARC Region, Autodesk.

Though GIS a multidisciplinary career, work experience and knowledge at different levels can be categorised under:

Digitisers / Executives / Operators: Fresh graduates from an IT background, with strong fundamental knowledge of GIS / CAD software, hardware like GIS Servers and geographical are expected for such a role. The job profile involves tasks like geo-spatial database editing and processing, Computer-aided Mapping and drafting by creating 2D and 3D environments, quality checks and analysis of existing databases and creating analytical presentations from spatial data available in figures and tables.

"A strong fundamental understanding of databases and database relations is necessary. It is necessary to know how to structure queries and execute them with an eye for detail, accuracy and quality. In data, each element is unique and needs special attention, whereas in programming, once you write a program everything comes out correct", informs Rohan Verma, founder and head of MapMy India.com.

Those interested in working on the GIS software development are required to learn programming languages like C, C++, C#, ADO.NET and Java, while those looking to work on the Web development side should know scripting languages (both server-side and client-side) like JavaScript, AJAX, ASP, PHP and Perl.

"The major tasks for beginners in Web Development involve using JavaScript (AJAX in particular) and interfacing with APIs such as Google Maps, Yahoo Maps, and other services", says Kaushal Gala, ex-GIS Web Developer currently working at Dell Inc. in the US.

At the beginner level, you can get to know the work cycle of GIS projects, and generally get exposed to the industry. You can super-specialise in a particular domain by working with organisations that offer GIS services in that area. After working for about two to three years, you advance to the level of Engineer / Technician.

Engineers / Technicians / Developers: Developers are expected to design and develop applications using GIS software engines and GIS servers to meet project requirements. These applications function with the database of the digitised files at the back end. Engineers and Technicians are required to keep the database servers in sync with the GIS applications being developed.

At this stage, they're expected to acquire comprehensive knowledge and experience of the usage as well as development of GIS software and tools.

Analysts: These are professionals who have strong fundamentals of mapping, are excellent with GIS applications, and have had some work experience as developers or technicians with GIS projects. Their job involves the study and analysis of GIS projects, and where necessary, suggest tools to meet the project's requirements.

GIS Analysts with enough experience can also choose to be independent consultants.

Project Co-ordinator/ Manager: After managing GIS projects for about five or more years, one can expect to be at the position of a GIS Project Co-ordinator/ GIS Manager. In GIS services providers, project managers play crucial role in getting more business. Usually, professionals stay at this position for 5-8 years but in exceptional cases, they manage to jump directly to the top level before that period.

Administrative Positions: Administrative positions like Business Development Manager involve interacting with clients, and bringing in more projects for the organisation. Their roles bend more towards the management side, but knowledge of GIS technologies and market trends—both domestic and international—is crucial.

Verma echoes, "New technologies are constantly coming, and this is a fast evolving field, requiring candidates to stay abreast with the latest happenings—they must take the initiative to stay up to speed if they want to keep growing at the pace of the industry."

System Requirements

Careers in GIS start with a passion for geography—ideally cultivated in school.

"Graduates from myriad fields such as civil engineering, geology, zoology, geo-technology, geo-informatics, botany, statistics, economics, commerce and business, geography including cartography, remote sensing, environmental science and sociology are eligible pursue career in GIS", says Mandar Mehta, CEO and founder of Datapoint Computer Services, which offers Engineering Services Outsourcing, KPO, BPO, ERP, Project Management, Digitization/CAD, GIS and e-Governance services.

Interested students from Computer Science and Engineering, Programming, Civil Engineering, CAD / CAM professionals, database systems specialists and Web technology specialists find their place in GIS software and Web development.

Points To Ponder

GIS is a growing industry, and there are initiatives that need to be taken to increase its popularity and fill the employment void.

"Along with awareness about GIS as mainstream and upcoming career, there is need to develop and follow a uniform academic curriculum at the university level. Increased support from Government for using GIS in all its departments for generating trained work force is expected, since GIS companies also look to the government for geospatial data", says Mehta.

There is also a need for including GIS-related topics at the secondary school level itself. "GIS and related subject courses are rarely available at the undergraduate level in India", notes Saha. Though many private institutes offer courses in GIS and related areas, the cost of training at these institutes is much higher compared to universities.

"Data conversion is the core business area currently with lot of opportunities arise due it's applications in multiple business domains."

Manideep Saha
Head-Infrastructure for India-SAARC region, Autodesk

Payday

Designation	Salary Per Annum (INR)	Experience (Years)
Administrative Positions	10 lakh and above	8–10
Project Lead/Project Manager	8–10 lakh	6–7
Analysts	6.4–8 lakh	3–5
Senior Designer	4 -6 lakh	1–3
Digitizers/Executives/Operators	1.2–3.2 lakh	-

Courses

In India, there are a number of courses that you can opt for: a B.Sc. / M.Sc. in GIS or Remote Sensing, an M. Tech in GIS or Geo-informatics, and so on. At the Master's level, students can choose electives to specialise their education. In addition, to several short-term courses like diplomas and certifications are also available. Many institutes—both Indian and international—even offer distance learning, some online. Do remember to thoroughly evaluate your university if you're pursuing an online education.

Even government institutions and organisations like the Indian Space Research Organisation (ISRO), the Geological Survey of India, the Space Application Centre at Ahmedabad, the National Remote Sensing Agency (NRSA), and the Survey Training Institute at Hyderabad offer short- and long-term courses, and also conduct workshops frequently for an affordable fee.

Internships

One of the most important and exciting parts of a career in GIS is the internship—real-world experience is a must. GIS employers are always looking for cheap labour for low- and assistant-level GIS tasks like data conversion. On the GIS software development side, interns could be given module programming, GIS software engine programming and documentation, or even trained as research and development assistants for GIS software. Students with internships with GIS companies obviously get preference over others, but having no experience is *not* a demerit. Companies are always open to employ interns, given the current dearth of a skilled workforce.

The Future

"Today, the structure and uniform spatial information about different regions of India isn't readily available as data sets with the Government, but this indicates immediate opportunities in the data conversion business, and also multiple opportunities in its application in several domains," says Mehta.

Though the pace of the GIS industry growth is slow right now, there is a great demand for skilled GIS professionals—not only for Government, but also for commercial businesses. Kapil Sibal, the Union Minister for Science and Technology, at the Map World Forum in January 2007, mentioned that the Indian GIS market is expected to shoot beyond the $1 billion mark by 2012. The estimated data conversion services exports were $100 million in 2007, and are expected to achieve $500 million in the next five years. The global geospatial industry growth rate ranged between 12 to 15 per cent annually.

Verma anticipates, "The future prospects of the Indian GIS industry are huge, because of the upcoming demand for premium quality digital maps and consumer navigation services. There will be demand at all levels of the value chain—from collecting GIS data in the field accurately, quickly and smartly, to compiling and processing that to create digital navigable maps, building products and services—both for consumers and enterprises—on top of the maps."

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